

Virtual conferencing

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Current multi-party video- and audioconferencing systems limit natural communications between participants. People communicate by speech, facial expressions and body gestures. In interactions between three or more people, these communications channels are directed towards particular participants. Spatial proximity and gaze direction are therefore important elements for effective conversational interactions, and yet are largely unsupported in existing conferencing tools. Advanced audioconferencing systems do simulate presence in a shared environment by using 'virtual humans' to represent the people taking part in a meeting, but the keyboard and mouse are used to direct conversations to specific people or to change the visual representation to simulate emotion.

This paper describes an experimental implementation of virtual conferencing, which uses machine vision to control a realistic virtual human, with the objective of making 'virtual meetings' more like physical ones. The computer vision system provides a more natural interface to the environment, while the realistic representation of users, with appropriate facial gestures and upper body movement, gives more natural visual feedback.

1. Introduction

Conferencing systems should enable remotely located groups or individuals to communicate effectively. Current video- and audioconferencing systems restrict the realism of that communication. Virtual conferencing uses a shared computer-generated space as an environment that more closely resembles a physical meeting, to overcome some of the barriers. Individuals are represented by realistic 'virtual humans' that move naturally.

Initially this paper explores the benefits of virtual conferencing, and outlines some of the essential requirements. It then focuses in detail on the underlying technology developed for an early virtual conferencing demonstration (see Fig 1). Finally, standards and future study areas are discussed.

2. Why virtual conferencing?

Current audio- and videoconferencing systems, while enabling effective communication at a distance, limit the naturalness of communication in the following ways.

- Audioconferencing allows multiple people to communicate using audio only. For larger conferences it is difficult to tell who is talking and there are no visual cues, such as body language and facial expressions, to complement the interaction.

- Videoconferencing adds visual information to the audioconference thereby reintroducing the visual cues for interaction. However, current systems are not able to create spatial positioning for multiple participants and therefore body language and conversations cannot be naturally directed to the relevant people in the conference. In addition, multiple video windows do not scale well both in terms of display and bandwidth.

Virtual conferencing, however, introduces the spatial cues into the communications environment, by representing participants as 'virtual humans' in a computer-generated space (Fig 1). Gestures and body language can be directed towards other people. Visual cues, such as who is speaking, are maintained by customising the visual appearance of each user's 'virtual human'.

Virtual conferencing provides additional advantages:

- the bandwidth required to support the environment is low, as, after initialisation, only changes in position of the 'virtual humans' are transmitted,
- a virtual conference can potentially support interactions between much larger groups of people, the terminals do not have to receive and process multiple video streams, and more 'virtual' space is available in the 3-dimensional virtual world than on 2-dimensional video windows,

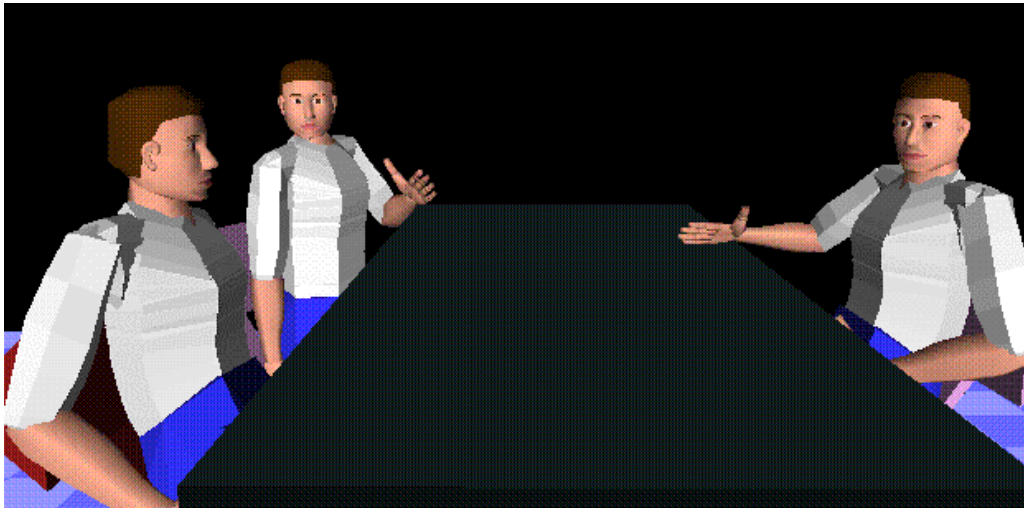


Fig 1 Virtual conference.

- the use of 'virtual humans' allows flexibility to tailor the environment to achieve the objective of the conference, the complexity of the virtual representations possibly varying dependent on the requirements of the interaction — if the number of people in the meeting is large, the detail of the users' representations may need to be reduced [1]; on the other hand, with a small number of participants it may be both appropriate and technically feasible to use more complex representations, revealing all body language and facial expressions in the interaction; if the meeting is focused on particular tasks or data in a group-working scenario, the detailed representation of users may prove to be relatively unimportant [2, 3]; in this case available computer resources can be targeted appropriately.

3. Requirements for creation and control of a virtual human

There are a number of fundamental requirements for the creation and control of virtual humans used in a conferencing environment:

- the virtual humans must be rendered in real time, enabling the users' expressions and movements to be replicated near-instantaneously — delays in simulating the gestures, expressions and body language can affect the meaning of the messages being communicated,
- the representations should be personalised to the users allowing the participants in the conference to readily recognise each other,
- the control of the virtual humans should be non-intrusive, so each participant can concentrate on the task, rather than on the underlying technology used to deliver the conferencing environment.

4. Face reading

The majority of virtual reality systems use data-gloves or headsets to track the movement of the user's hands and head. These methods for controlling 'presence' [2] within a computer-generated environment are adequate when the user is interacting with data, but are restrictive when the primary objective is communication with other people. Research at BT Laboratories (BTL) is looking at how these problems may be overcome using computer vision-tracking systems.

A vision system [4] has been constructed to analyse facial details without recourse to specialist image processing hardware. This system consists of a PC containing a simple video input card, which is connected to a video camera aligned to view the user's head and shoulders. Images from this camera are analysed to find the main facial features such as eyes, mouth and eyebrows, and characterise facial expressions and head orientation.

In the following sections, the techniques used to locate the facial features are described.

4.1 Locating the eyes

The first stage in the facial feature location is an eye search. Once the eyes have been located, the detection of other facial features is made easier.

The eyes are located by scanning model images of left and right eyes over the top centre region of each video frame until a high correlation is found. After the first few frames, the search time is reduced by constraining the search to the areas immediately surrounding each eye. In the event of failure, the area is widened for each frame until a match is again found.

If multiple matches are found, due to background features with a similarity to an eye template, then a score is derived from the horizontal angle, correlation coefficients and separation. The features with the highest scores are then used.

4.2 Head orientation

Knowing the positions of the eyes in relation to the head boundaries in an image dictates where a face is pointing. The skin and hair colours are used to segment the face from the background. Figure 2 shows the resultant location of the eyes and the segmentation of the sides and top of the head.

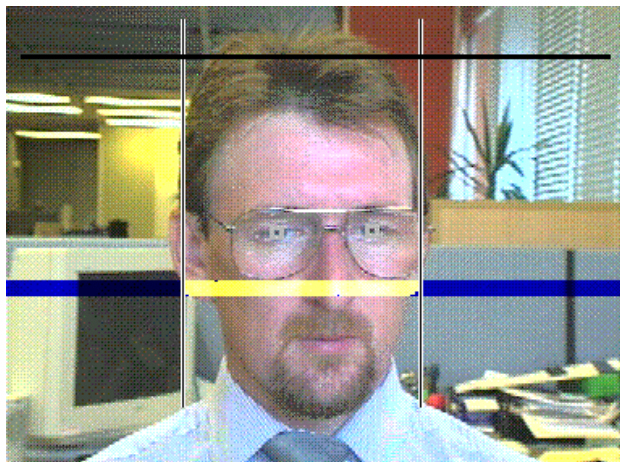


Fig 2 Locating the eyes and the extremities of the head.

During the first few frames, after eye location, a skin image sample is taken from just below the eyes, and hair from above. The colour levels red, green, blue (RGB) of each pixel from these samples are used to calculate an average skin and hair colour. The Euclidean distance from the mean RGB distribution to each pixel to be classified gives the closeness to skin or hair. The sides of the head are determined by finding the ends of the skin class along the horizontal stripe and similarly with the vertical.

The distance of the head from the camera is derived from the ratio of the number of pixels across the head to the assumed head diameter. Simple geometry is used to estimate the face pose angle.

4.3 The mouth

Knowing eye position and head width allows a good estimate of mouth location (see Fig 3). Greater accuracy is achieved by passing a horizontal edge detector over the mouth region revealing the mouth as a set of long horizontal lines. This works even in the presence of facial hair which tends to lie vertically.

The horizontal edge detector is a simple image processing function where a mask, three pixels square, is scanned over the image. The mask contains the pixel values

shown in Fig 4. Each pixel in the mask is multiplied by the pixel in the image which it overlays. If the sum of these values is greater than a predefined threshold, the centre pixel is classified as an edge pixel.

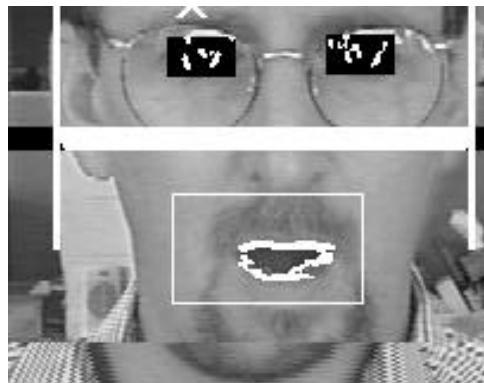


Fig 3 Mouth tracking.

-2	-2	-2
0	0	0
2	2	2

Fig 4 Pixel mask.

The overall width and height of the mouth are normalised and transmitted. These values are used as weighting factors in the mouth animation. For conferencing applications, the mouth shape parameters need to be transmitted at more than 15 frames/sec for a good representation of speech.

The shape of the upper lip is used to determine the degree of smiling to apply to the mouth model. This is calculated using the vertical position of the centre of the upper lip in relation to the vertical position of the corners of the mouth. The resting shape of the mouth is determined when the face reader is initialised. The resting shape is subtracted from all subsequently calculated positions.

4.4 Eyebrows

The eyebrows are located in the images using template matching in a similar manner to that for the eye location. The resting positions of the eyebrows are measured and used as an offset to any position change. At present, some of the subtler eyebrow movements have proved difficult to detect, although frowning is detected by counting the number of vertical lines between the eyebrows. A mapping from eyebrow position to muscle action is performed, giving appropriate weighting factors for the frontalis and corrugator muscles.

5. Body tracking

Upper body gestures and movement are important for effective communication. The ability to gesture in free space enhances the 'naturalness' of the interaction.

Determining the position of the hands and the head in 3-D involves two cameras in the stereoscopic vision system developed by MIT Media Lab, and known as STIVE [5]. The heads and hands are segmented from the background for each of the camera views using skin colour (see Fig 5). The 3-D position and orientation of the head and hands are determined through geometric computation given the information from the right and left camera views.



Fig 5 STIVE body-tracking system.

Person-independent tracking is enabled by training the system to recognise skin colour across a number of users. Images of different users are captured initially and examples of skin colour presented to the calibration system, by clicking on the heads and hands within the images.

6. Creating realistic virtual humans

The previous section focused on extracting the information required to control a virtual human using non-intrusive tracking systems, and this section goes on to consider the creation of 'virtual humans' that represent the participants in a conference. The virtual human computer model is a three-dimensional mesh of polygons, as illustrated in Fig 6.

The complexity of the 3-D computer model required to represent a user in a shared environment is dependent on the application and the nature of the information which needs to be communicated. In applications where visual feedback is not required then the complexity of the model can be reduced significantly and may be used merely to indicate the presence of a user. In interactions where the focus of attention is directly on the other users in the interaction rather than on a common task, the facial expressions and body language become paramount for providing realistic

and natural interaction. Here the interest is in generating realistic 'human-like' representations of users while maintaining the real-time manipulation of these models to synthesise natural movement.

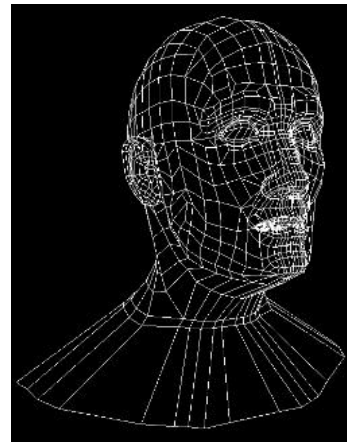


Fig 6 Polygonal head model.

7. Head modelling

Initially the head model is a generic head, which, in a process referred to as conformance, is modified semi-automatically to match the shape of the user's head. Images of the head are captured from the front, both sides and back. Additional images of closed eyes, teeth and the tongue are also captured to aid the realism in the final model. The range of images captured is shown in Fig 7.



Fig 7 High-quality images of the user.

View-based recognition techniques are used to detect significant features such as the eyes, nose and mouth [6]. The positions of other features are provided within the front image and one of the side images. The polygonal mesh is conformed to match the significant facial features of the user, such as distance between the eyes, width and height of

the head and the distance of the eyes to the mouth. Once a model is produced, the head is textured with the captured images without the need to stretch regions of the images, as the underlying model is of the appropriate size and shape.

This is achieved through one of two methods — manually in a modelling package or using an automated texturing system [7].

7.1 Manual texturing

The combined texture map together with the head model, conformed to the size and shape of the user's head, is loaded into 'Medit', a 3-D modelling package [8].

The user can then manipulate the texture map by hand so that it maps correctly on to the model. Medit will then export the textured head model in an appropriate format.

The resulting 'normal' textured head is shown in Fig 8.



Fig 8 'Normal' head resulting from manual texturing.

7.2 Automatic texturing

Automatic texturing applies the photographs of a subject taken from different aspects (Fig 7) to a conformed head. The algorithm applies regions of each photograph to different areas of the head model, so that maximum image resolution is maintained at all times.

For optimum results, a colour equalisation and blending of the photographs should be performed before automatic texturing. This will correct any lighting differences when the photographs were taken and ensure the texture blends seamlessly once applied to the head model. A resulting head textured automatically is shown in Fig 9.

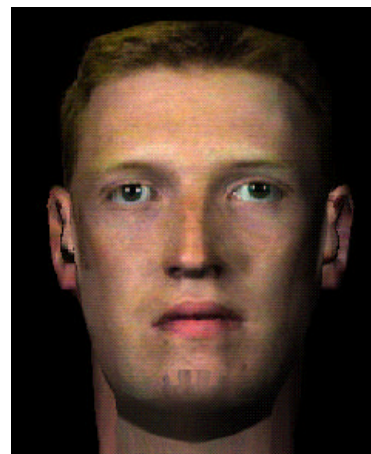


Fig 9 Head textured automatically.

8. Mouth modelling

The next stage of the modelling process is to define mouth models, which sufficiently describe the extremes of mouth shapes produced during normal speech. Initially the mouth shapes are modelled for generic head model using video sequences of speech as references. The scaling transformations applied to the generic model for each user are then applied to the generic set of mouth models to produce the user-specific set of models.

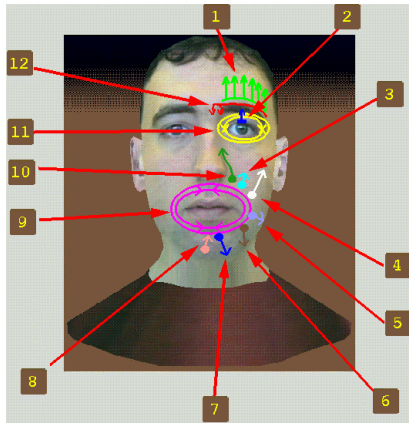
9. Facial expressions

The synthesis of facial expressions is achieved by modelling the action of a number of muscles within the face. To simplify this problem only the major muscles within the face have been modelled. These muscles are shown in Fig 10.

The effect of each muscle on the polygonal mesh has been modelled by hand for both the relaxed state and at maximum contraction. A weighting factor, representing the contraction of each muscle in the range 0 to 1, is set. An intermediate mesh for each muscle is generated, assuming each muscle acts linearly from relaxation to maximum contraction. The resulting meshes are then combined linearly to produce a single resultant expression. In this way, a whole range of facial expressions can be produced by combining the action of several muscles. Again, to make the facial expressions user-specific, the conformation transformations are applied to each of the muscle models.

10. Body modelling

Up to this point, the paper has concentrated on the modelling of the face and facial expressions. A realistic representation of a human in a virtual environment requires a body that enables upper-body gestures. To this end, the JACK human body modelling software [9] has been integrated with the facial modelling work.



- 1 Frontalis — raises the eyebrows, as in surprise.
- 2 Eyelid Muscle — opens and closes the upper eyelids.
- 3 Upper Lip Muscles — raises the upper lip to produce a sneer.
- 4 Smiling Muscle — raises the corners of the mouth towards the cheekbones.
- 5 Risorius — stretches the lower lip horizontally into a grimace.
- 6 Triangularis — the ‘frowning muscle’ which lowers the corners of the mouth.
- 7 Lower Lip Muscle — pulls down on the lower lip, as in pronouncing ‘s’.
- 8 Mentalis — the ‘pouting muscle,’ which raises the chin and lower lip.
- 9 Lip Purser — constricts the mouth, as in kissing or pronouncing ‘oo’.
- 10 Nose — Upper Lip Muscle — flares the nostrils and pulls the upper lip to produce a wince.
- 11 Squinting Muscle — compresses the skin around the eyes.
- 12 Currugator — furrows the brow by pulling it downward.
- 13 Jaw Muscles — open and close the mouth.

Fig 10 Muscles within the face.

The JACK body model consists of joints modelled closely on that of a real human, so that the appropriate degrees of freedom and joint limits are expressed in its movement. The software is primarily used in the design industry, in particular by automotive manufacturers for human factors engineering, where they require virtual humans of the appropriate size and shape, which move naturally.

The JACK software environment is a modular system giving users the ability to extend the basic functionality. This feature is used to integrate the facial modelling work with the body model. The standard JACK human model is hierarchical in the sense that the human object is made up of a number of segments, which describe the visual appearance of the human and a number of joints which define how the segments are attached to each other. The head segment of the standard JACK virtual human is replaced with the new head segment modelled previously. As the new segment is placed in the same hierarchical location within the human object description of the JACK virtual human, all the joints defined regarding the connection to the head still apply — only the visual appearance of the head is changing.

The integration of the face model with the body model produces a recognisable virtual human with natural upper body and facial movement as shown in Fig 11.

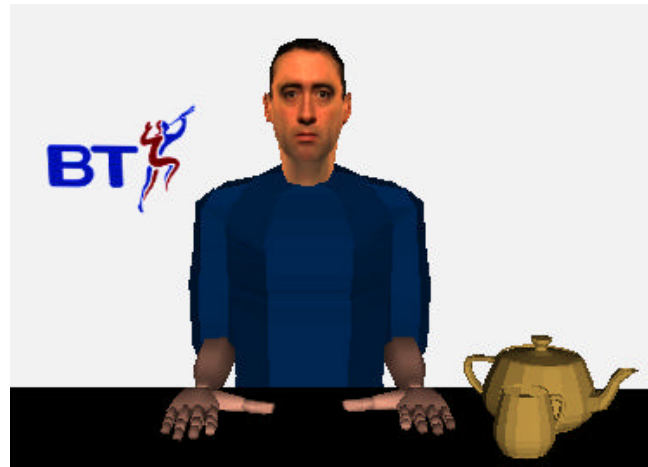


Fig 11 Virtual human.

11. Natural movement

To produce realistic visual feedback in an application, the user's representation must move naturally, not only with large body motions but also with finer facial movements. In addition, the user's speech signal should be synchronised with the virtual human's mouth and facial movements. To synchronise the audio with the animation of the facial movements frame-based simulation system software [10] is used. This enables the computer system to maintain a fixed simulation frame rate, even if complex changes to the virtual environment are apparent.

11.1 Speech synchronisation

The animation of the mouth is achieved by morphing between three different mouth shapes — mouth closed, mouth open, as in ‘ah’, and lips pursed, as in ‘oo’. The face-reading system monitors the subject and determines directly the weightings for each shape. Natural movement of the face is limited by the ability to track movement of facial features in real time and with sufficient resolution.

In a complementary approach for computer-generated speech, BT's text-to-speech system, Laureate [11], has been used to control the movement of the mouth with much greater accuracy. This is achieved using a linear interpolation algorithm to morph between the complete range of mouth shapes. The algorithm extracts a stream of mouth shapes and the duration each is active from the text-to-speech system. This is important as different phonemes take differing times to be pronounced. Given this information the algorithm calculates the appropriate interpolated mouth shape at each frame boundary, given the system frame rate, and renders the appropriate model in real

time (see Fig 12). In addition, the appropriate portion of the speech stream is output.

All calculations and rendering of the new model need to be performed within one frame duration. The system frame rate is therefore dependent on the processing power of the machine, and a frame rate of 24 Hz is achievable on a graphical workstation.



Fig 12 Interpolation between 'H' and 'L' mouth shapes.

11.2 Head rotation

Head rotations (Fig 13), are achieved by multiplying the vertices in the head by a rotation matrix. This rotation is gradually reduced in the neck region giving the impression that the neck is twisting. The angle of rotation about the X and Y axes are given by the face-reading tracking system, with new rotations being rendered each frame.



Fig 13 Head rotations.

11.3 Facial expressions

Expression changes are currently achieved through the variation of each of the muscle's weighting factors, which are calculated on a frame-by-frame basis. Currently the face-reading system provides weighting factors for a limited number of the muscles. These are the frontalis and corrugator muscles, which affect the eyebrows, and the 'smiling' muscle.

11.4 Body movement

Upper body movement is achieved using inverse kinematics, which is part of the JACK body-modelling software. Inverse kinematics is the technique used to calculate the position of a limb given its end position. A

constraint chain, specifying which joints to position automatically given the position of a constraint, is set up for the left and right arms. The respective starting joints are the left and right shoulders and a hold constraint is specified at the right and left hand palm centre. The xyz position of the left and right hand are obtained for each frame through a remote procedure call to the body-tracking system. This information is specified as the global position of the hold constraints. The inverse kinematics algorithm then solves for the positions of the joints within the chain on an iterative basis. The solution provides a realistic simulation of arm movement.

Since there is no direct control of the elbow joint there are a number of joint positions, which solve the criteria of the algorithm. Therefore, it is possible for the elbow to appear in an unusual position. To minimise the problem the wrist joint is frozen, reducing the number of variables within the joint chain.

Movement of the upper body is controlled in a similar fashion, with the joint chain specified from the waist to the top of the neck. The performance of the inverse kinematics algorithm is optimised by reducing the time limit for the algorithm solution. This reduces the accuracy of the final position of the hands and head but increases the frame rate, thereby optimising the lag between movements in the real and virtual world.

12. Virtual conferencing scenario

The preceding sections have described the underlying technology for delivering a virtual conference. Each user appears as a realistic representation of themselves within a computer-generated environment, with computer vision systems to track facial expressions and body movements. As people interact, the other participants see appropriate body postures, gestures and facial expressions. Figure 14 summarises the set-up for an individual user.

The set-up shown in Fig 14 was implemented, as a demonstration, to determine the performance of the system and to gauge initial reaction to the use of 'virtual humans' in conferencing.

The system performed well, achieving a frame rate of 8 frames/sec for the movement of the 'virtual human'. The tracking systems can run at faster rates, as stated earlier, but system performance is limited by the rendering of the 'virtual human'.

The data sent across the network, to control the 'virtual human', was very small. Changes in position to the facial muscles, head movement and body movement in addition to the speech were all that were communicated. The data rate to control the 'virtual human' excluding speech was less than 8 kbit/s. No effort was made to compress the data.

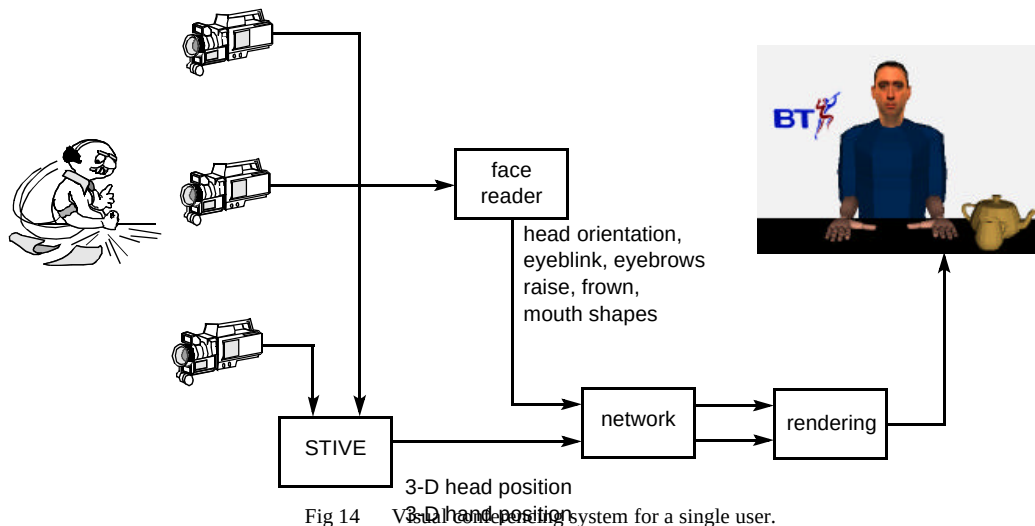


Fig 14 Visual conferencing system for a single user.

13. Discussion

The use of 'virtual humans' as part of a conferencing application is an area of increasing research attention. A number of universities [12, 13] are developing realistic 'virtual humans', while other research groups [14, 15] are developing similar conferencing scenarios which combine computer vision systems with 'virtual humans'. It is therefore important that consideration is given to the standardisation of the interfaces to the 'virtual human', so that future conferencing systems can make use of this technology.

The pre-eminent standard looking at this area is the Synthetic/Natural Hybrid Coding (SNHC) section of the MPEG 4 standard [16]. The synthetic coding currently under standardisation includes media integration of text and graphics, face and body animation, texture coding and static and dynamic mesh coding.

The face and body animation working group is considering both the appearance and animation of the face and body. The appearance of the virtual human is defined using facial and body definition parameters (FDPs and BDPs), while facial and body animation parameters (FAPs and BAPs) define the movement of the 'virtual human'. The definition parameters are used to customise a generic head and body model to a particular person.

The facial animation parameters are based on the study of minimal facial actions and are closely related to muscle actions. They represent a complete set of basic facial actions and therefore allow the representation of most facial expressions. The body animation parameters describe the movement of the 'virtual human' using joint angles.

A more detailed discussion of the MPEG 4 standard is given in Whybray et al [17].

The question of customer acceptability of such systems is also under active investigation. The development of the MPEG 4 standard will lead to synthetic data appearing in future conferencing systems, while the work described here will allow the suitability of this new technology to be evaluated. As computer-processing power increases, the realism of the representations will improve, and the boundary between real and virtual data will become blurred. The question is how real do the 'virtual humans' need to become before customers feel as though they are interacting naturally? Reaction to the demonstration was that the head was very realistic, but the simple visual appearance of the body detracted from the realism of the 'virtual human' as a whole.

Early subjective experiments have been carried out by ATR Communication Systems Research Laboratories in Japan, using a similar virtual teleconferencing system [18]. The experiments focused on the required complexity of the 'virtual humans' and the naturalness of the movements. The results show that there is an optimum level of complexity of 'virtual human', above which there is no significant increase in perceived quality. The research team concluded that further improvements in smoothness, naturalness and speed of movements are required before this technique becomes acceptable for conferencing.

These studies have highlighted a number of important areas for future work, including:

- the rendering of the head was much more realistic than that of the body, giving conflicting messages as to whether the virtual human was a cartoon animation or a video-like representation,
- the complexity of the 'virtual human' needs to be balanced with the speed and smoothness of movements,

- occasional gross errors in the body- and face-tracking systems need to be almost eliminated in order to make the gestures appear natural,
- subtle movements in the eye region are very important in the replication of body language — this cannot be animated using the same methods that have been described for the mouth, and a technique such as video texturing may be more appropriate for the eye regions.

In addition to such challenges in image processing and rendering, much work also remains on issues of interface, application design and usability before commercial services can be developed.

14. Conclusions

The ability to model humans and control their movement in a virtual environment without the use of sensors or keyboards has been demonstrated. The BTL system used a single virtual human and image processing system, and demonstrated several of the essential requirements for the creation and control of virtual humans for real-time conferencing.

Virtual conferencing promises scalability and increased presence in multi-point meetings. It enables appropriate spatial proximity, social cues and gestures to be introduced into multi-location conferences, and with appropriate coding can reduce the bandwidth requirements. In a future of diverse, multimedia telepresence services, the virtual human is certain to be an important contributor.

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Andy Mortlock graduated from Portsmouth Polytechnic in 1989 with a BEng in Electrical and Electronic Engineering. He joined BT Laboratories after graduation where he worked on the development of equipment to test the speech transmission performance of ISDN terminals and non-intrusive measurement of international telephony networks. His research has moved on to exploring the use of virtual reality for teleconferencing. In particular he has focused on the creation of realistic 3-D talking heads and 'virtual humans'.



Dave Machin gained a degree from the University of Bradford in 1975, joining BT Laboratories to work on mathematical modelling of transistors for integrated circuits. After a number of years in this area he spent five years on a collaborative European project for automatic design validation of integrated circuits using electron beams. In 1990, he joined a team working on subjective testing of telephony services and developed a tool using digital signal processors to emulate international telephony networks. He has recently moved into the area of machine vision for teleconferencing.



Stephen McConnell gained a BEng (Hons) degree in Electronic System and Control Engineering from Sheffield Hallam University in 1993.

He joined BT Laboratories in September 1994. Initially, he was involved in research on 2-D talking heads which led on to research on 3-D talking heads.

Currently his main research interest is in the application of realistic 'virtual humans' for virtual conferencing and as avatars in shared virtual spaces.



Phil Sheppard graduated from Birmingham University in 1986 with a BSc (Hons) in Computer Science and Electronic Engineering. He joined BT Laboratories on graduation where he has worked on ISDN transmission over copper, speech coding and recognition, network emulation, algorithm development for customer opinion modelling and non-intrusive measurement of voice networks for which he is an ITU-T Rapporteur. He currently leads a group working on a diverse set of projects to provide natural communications systems — including research into non-intrusive

measurement techniques, conferencing systems (including VR technology and video image processing) and development of BT's 3-D talking head. He gained an MSc in Telecommunications Engineering from the University of London in February 1997.